Skills England

Machine learning engineer

Key information

Reference: ST1398
Version: 1.0
Level: 6
Typical duration: 24 months
Typical assessment period: 4 months
Minimum hours for compliance: 418
Route: Digital
Integration: None
Maximum funding: £22000

Date updated: 09/06/2026
Approved for delivery: 18 December 2024
Lars code: 795
EQA provider: Ofqual
Example progression routes:
Artificial intelligence (AI) data specialist
Review: this apprenticeship will be reviewed in accordance with our change request policy.

Apprenticeship summary | Occupational standard | EPA plan

Contents

End-point assessment plan

V1.0

Introduction and overview

This document explains the requirements for end-point assessment (EPA) for the machine learning engineer apprenticeship. End-point assessment organisations (EPAOs) must follow this when designing and delivering the EPA.

Machine learning engineer apprentices, their employers and training providers should read this document.

A full-time machine learning engineer apprentice typically spends 24 months on-programme. The apprentice must spend at least 12 months on-programme and complete the required amount of off-the-job training in line with the apprenticeship funding rules.

The EPA should be completed within an EPA period lasting typically 4 months.

The apprentice must complete their training and meet the gateway requirements before starting their EPA. The EPA will assess occupational competence.

An approved EPAO must conduct the EPA for this apprenticeship. Employers must work with the training provider to select an approved EPAO.

This EPA has 2 assessment methods.

The grades available for each assessment method are below.

Assessment method 1 - project evaluation report, presentation and questioning :

- fail
- pass
- distinction

Assessment method 2 - professional discussion :

- fail
- pass
- distinction

The result from each assessment method is combined to decide the overall apprenticeship grade. The following grades are available for the apprenticeship:

- fail
- pass
- merit
- distinction

EPA summary table

On-programme - typically 24 months	The apprentice must: • complete training to develop the knowledge, skills and behaviours (KSBs) outlined in this apprenticeship's standard • complete training towards English and mathematics qualifications in line with the apprenticeship funding rules
End-point assessment gateway	The apprentice's employer must be content that the apprentice is occupationally competent. The apprentice must: • confirm they are ready to take the EPA • have achieved English and mathematics qualifications in line with the apprenticeship funding rules For the project evaluation report, presentation and questioning , the apprentice must submit a project brief. To ensure the project allows the apprentice to meet the KSBs mapped to this assessment method to the highest available grade, the EPAO should sign-off the project's title and scope at the gateway to confirm it is suitable. A brief project summary must be submitted to the EPAO. It should be no more than 500 words. This needs to show that the project will provide the opportunity for the apprentice to cover the KSBs mapped to this assessment method. It is not assessed. Gateway evidence must be submitted to the EPAO, along with any organisation specific policies and procedures requested by the EPAO.
End-point assessment - typically 4 months	**The grades available for each assessment method are below** Project evaluation report, presentation and questioning : • fail • pass • distinction

	Professional discussion : • fail • pass • distinction **Overall EPA and apprenticeship can be graded:** • fail • pass • merit • distinction
Professional recognition	This apprenticeship aligns with: • BCS, The Chartered Institute for IT for Advanced RITTech
Re-sits and re-takes	• re-take and re-sit grade cap: pass • re-sit timeframe: typically 3 months • re-take timeframe: typically 6 months

Duration of end-point assessment period

The EPA is taken in the EPA period. The EPA period starts when the EPAO confirms the gateway requirements have been met and is typically 4 months.

The EPAO should confirm the gateway requirements have been met and start the EPA as quickly as possible.

EPA gateway

The apprentice's employer must be content that the apprentice is occupationally competent. That is, they are deemed to be working at or above the level set out in the apprenticeship standard and ready to undertake the EPA. The employer may take advice from the apprentice's training provider, but the employer must make the decision. The apprentice will then enter the gateway.

The apprentice must meet the gateway requirements before starting their EPA.

They must:

- confirm they are ready to take the EPA
- have achieved English and mathematics qualifications in line with the apprenticeship funding rules
- submit a project brief for the project evaluation report, presentation and questioning

The apprentice must produce a project brief .

Gateway evidence must be submitted to the EPAO, along with any organisation specific policies and procedures requested by the EPAO.

Order of assessment methods

The assessment methods can be delivered in any order.

The result of one assessment method does not need to be known before starting the next.

Project evaluation report, presentation and questioning

Overview

The project report assessment method involves the apprentice completing a significant and defined piece of work that has a real business application and benefit. This process may include for example, research, analysis and the completion of tasks or activities to achieve the outcome. The assessment method will have an output at the end of the defined piece of work. The work completed for the project report assessment method must meet the needs of the employer's business and be relevant to the apprentice's occupation and apprenticeship.

This assessment method has 2 components:

- completion of the defined piece of work for the project with a project output
- presentation with questions and answers

Together, these components give the apprentice the opportunity to demonstrate the KSBs mapped to this assessment method. They are assessed by an independent assessor.

Rationale

This assessment method is being used because:

A project evaluation report is the most valid method as it allows the demonstration of professional competence.

The project is based on a real life example of the apprentices' everyday work in their industry.

Therefore, ensuring that they can demonstrate the KSBs in practice.

Producing a project evaluation report and presentation reflects normal professional practice, so this assessment method is appropriate.

Apprentices are required to be concise and precise in their use of language in written and verbal communication.

The project evaluation report offers a realistic opportunity to combine project management, examples of data products and formal writing enabling the apprentice to reflect on approaches taken.

It is a holistic assessment method, allowing the apprentice to demonstrate KSBs in an integrated way.

By writing the evaluation report on the project and being questioned to understand rationale for choices made, risks and problems identified, resolutions and areas where further action could be required.

This method will enable the apprentice to showcase their professional competency. The project is completed before gateway and is not graded.

The apprentice must start the project evaluation report after gateway. As the project evaluation report is assessed it must be completed after gateway.

Delivery

The apprentice must complete a project report based on any of the following:

a specific problem

a recurring issue

an idea or opportunity

Suggested projects include:

scaling of an existing system or process

making improvements to a current system

enhancing organisational capability

improvements to enhance environmental sustainability

To ensure the project report allows the apprentice to meet the KSBs mapped to this assessment method to the highest available grade, the EPAO must sign-off the project's title and scope at the gateway to confirm it is suitable. The EPAO must refer to the grading descriptors to ensure that projects are pitched appropriately.

The project output must be in the form of a report and presentation.

The apprentice must start the project report after the gateway. The employer should ensure the apprentice has the time and resources, within the project period, to plan and complete their project report.

The apprentice may work as part of a team to complete the project, which could include internal colleagues or technical experts. The apprentice must however, complete their project report and presentation unaided and they must be reflective of their own role and contribution. The apprentice and their employer must confirm this when the report and any presentation materials are submitted.

The apprentice may choose to end the assessment method early. The apprentice must be confident they have demonstrated competence against the assessment requirements for the assessment method. The independent assessor or EPAO must ensure the apprentice is fully aware of all assessment requirements. The independent assessor or EPAO cannot suggest or choose to end any assessment methods early except in the event of a health and safety, medical or safeguarding event. The EPAO is responsible for ensuring the apprentice understands the implications of ending an assessment early if they choose to do so. The independent assessor may suggest the assessment continues. The independent assessor must document the apprentice's request to end any assessment early.

Component 1: Project report

The report must include at least:

- an executive summary (or abstract)
- an introduction
- the scope of the project (including key performance indicators, aims and objectives)
- a project plan
- research outcomes
- data analysis outcomes
- project outcomes
- discussion of findings
- recommendations and conclusions
- references
- appendix containing mapping of KSBs to the report.

The report must also include:

Why a specific model was deployed for a certain data set

How the apprentice overcame challenge and made improvements as a result

How data bias was handled.

The project report will typically have a word count of 5000 words. A tolerance of 10% above or below is allowed at the apprentice's discretion. Appendices, references and diagrams are not included in this total. The apprentice must produce and include a mapping in an appendix, showing how the report evidences the KSBs mapped to this assessment method.

The apprentice must complete and submit the report and any presentation materials to the EPAO by the end of week 8 of the EPA period.

Component 2: Presentation with questions

The presentation with questions must be structured to give the apprentice the opportunity to demonstrate the KSBs mapped to this assessment method to the highest available grade.

The apprentice must prepare and deliver a presentation to an independent assessor. After the presentation, the independent assessor must ask the apprentice questions about their project, report and presentation.

The presentation should cover:

- an overview of the project
- the project scope (including key performance indicators)
- summary of actions undertaken by the apprentice
- project outcomes and how these were achieved
- how the apprentice overcame any challenges

The presentation with questions must last 50 minutes. This will typically include a presentation of 20 minutes and questioning lasting 30 minutes. The independent assessor must use the full time available for questioning. The independent assessor can increase the time of the presentation and questioning by up to 10%. This time is to allow the apprentice to complete their last point or respond to a question if necessary.

The apprentice may choose to end the assessment method early. The apprentice must be confident they have demonstrated competence against the assessment requirements for the assessment method. The independent assessor or EPAO must ensure the apprentice is fully aware of all assessment requirements. The independent assessor or EPAO cannot suggest or choose to end the assessment methods early, unless in an emergency. The EPAO is responsible for ensuring the apprentice understands the implications of ending an assessment early if they choose to do so. The independent assessor may suggest the assessment continues. The independent assessor must document the apprentice's request to end the assessment early.

The independent assessor must ask at least 6 questions. They must use the questions from the EPAO's question bank or create their own questions in line with the EPAO's training. Follow up questions are allowed where clarification is required.

The purpose of the independent assessor's questions is:

- to verify that the activity was completed by the apprentice
- to seek clarification where required
- to assess those KSBs that the apprentice did not have the opportunity to demonstrate with the report, although these should be kept to a minimum
- to assess level of competence against the grading descriptors

The apprentice must submit any presentation materials to the EPAO at the same time as the report - by the end of week 8 of the EPA period. The apprentice must notify the EPAO, at that point, of any technical requirements for the presentation.

During the presentation, the apprentice must have access to:

- audio-visual presentation equipment
- flip chart and writing and drawing materials
- computer

The independent assessor must have at least 2 weeks to review the project report and any presentation materials, to allow them to prepare questions.

The apprentice must be given at least 2 weeks' notice of the presentation with questions.

Assessment decision

The independent assessor must make the grading decision. They must assess the project components holistically when deciding the grade.

The independent assessor must keep accurate records of the assessment. They must record:

- the KSBs demonstrated in the report and presentation with questions
- the apprentice's answers to questions
- the grade achieved

Assessment location

The presentation with questions must take place in a suitable venue selected by the EPAO for example, the EPAO's or employer's premises. It should take place in a quiet room, free from distractions and influence.

The presentation with questions can be conducted by video conferencing. The EPAO must have processes in place to verify the identity of the apprentice and ensure the apprentice is not being aided.

Question and resource development

The EPAO must develop a purpose-built assessment specification and question bank. It is recommended this is done in consultation with employers of this occupation. The EPAO must maintain the security and confidentiality of EPA materials when consulting with employers. The assessment specification and question bank must be reviewed at least once a year to ensure they remain fit-for-purpose.

The assessment specification must be relevant to the occupation and demonstrate how to assess the KSBs mapped to this assessment method. The EPAO must ensure that questions are refined and developed to a high standard. The questions must be unpredictable. A question bank of sufficient size will support this.

The EPAO must ensure that the apprentice has a different set of questions in the case of re-sits or re-takes.

EPAO must produce the following materials to support the project:

- independent assessor EPA materials which include:
  - training materials

- administration materials
- moderation and standardisation materials
- guidance materials
- grading guidance
- question bank

- EPA guidance for the apprentice and the employer

The EPAO must ensure that the EPA materials are subject to quality assurance procedures including standardisation and moderation.

Professional discussion

Overview

In the professional discussion, an independent assessor and apprentice have a formal two-way conversation. It gives the apprentice the opportunity to demonstrate the KSBs mapped to this assessment method.

Rationale

This assessment method is being used because:

- It provides the apprentice with the opportunity to discuss and show case their depth of understanding the knowledge, skills and behaviours that may not naturally occur as part of the project.
- It allows the independent assessor to consider the context and sector that the apprentice operates within, giving flexibility to ensure that all the KSBs can be assessed appropriately.
- The professional discussion is cost effective, and it allows consideration of the potential need to conduct the EPA remotely.

Delivery

The professional discussion must be structured to give the apprentice the opportunity to demonstrate the KSBs mapped to this assessment method to the highest available grade.

An independent assessor must conduct and assess the professional discussion.

The following topics will be covered in the professional discussion:

Model scoping

Experiment and tracking

Model deployment

Collaborative working

Sustainability

Engineering principles

Model testing and improvement

Model management

Compliance and assurance

Collaborative working

Continuous professional development

The EPAO must give an apprentice 2 weeks' notice of the professional discussion.

The professional discussion must last for 90 minutes. The independent assessor can increase the time of the professional discussion by up to 10%. This time is to allow the apprentice to respond to a question if necessary.

The apprentice may choose to end the assessment method early. The apprentice must be confident they have demonstrated competence against the assessment requirements for the assessment method. The independent assessor or EPAO must ensure the apprentice is fully aware of all assessment requirements. The independent assessor or EPAO cannot suggest or choose to end the assessment methods early, unless in an emergency. The EPAO is responsible for ensuring the apprentice understands the implications of ending an assessment early if they choose to do so. The independent assessor may suggest the assessment continues. The independent assessor must document the apprentice's request to end the assessment early.

The independent assessor must ask at least 6 questions. The independent assessor must use the questions from the EPAO's question bank or create their own questions in line with the EPAO's training. Follow-up questions are allowed where clarification is required.

The independent assessor must make the grading decision.

The independent assessor must keep accurate records of the assessment. They must record:

- the apprentice's answers to questions
- the KSBs demonstrated in answers to questions
- the grade achieved

Assessment location

The professional discussion must take place in a suitable venue selected by the EPAO for example, the EPAO's or employer's premises.

The professional discussion can be conducted by video conferencing. The EPAO must have processes in place to verify the identity of the apprentice and ensure the apprentice is not being aided.

The professional discussion should take place in a quiet room, free from distractions and influence.

Question and resource development

The EPAO must develop a purpose-built assessment specification and question bank. It is recommended this is done in consultation with employers of this occupation. The EPAO must maintain the security and confidentiality of EPA materials when consulting with employers. The assessment specification and question bank must be reviewed at least once a year to ensure they remain fit-for-purpose.

The assessment specification must be relevant to the occupation and demonstrate how to assess the KSBs mapped to this assessment method. The EPAO must ensure that questions are refined and developed to a high standard. The questions must be unpredictable. A question bank of sufficient size will support this.

The EPAO must ensure that the apprentice has a different set of questions in the case of re-sits or re-takes.

The EPAO must produce the following materials to support the professional discussion :

- independent assessor assessment materials which include:
    - training materials
    - administration materials
    - moderation and standardisation materials
    - guidance materials
    - grading guidance
    - question bank
- EPA guidance for the apprentice and the employer

The EPAO must ensure that the EPA materials are subject to quality assurance procedures including standardisation and moderation.

Grading

Project evaluation report, presentation and questioning

Fail - does not meet pass criteria

THEME KSBS	PASS APPRENTICES MUST DEMONSTRATE ALL OF THE PASS DESCRIPTORS	DISTINCTION APPRENTICES MUST DEMONSTRATE ALL OF THE PASS DESCRIPTORS AND ALL OF THE DISTINCTION DESCRIPTORS
Model scoping K2 K3 K6 K7 S2 S4 S21	Outlines the stages of the machine learning lifecycle. K2 Explains their use of performance metrics when scoping machine learning engineering solutions by translating business needs and technical problems K7, S2 Explains how they apply project management methodologies and techniques for the machine learning activities. S4 Describes the risks of deploying new methods and models and how they assess system vulnerabilities, mitigating the threats or risks to assets and data bias. K3, K6, S21	Evaluates the impact of performance metrics on scoping machine learning engineering solutions when meeting business needs and technical problems. K7, S2
Experiment and tracking K8 K9 K10 K12 K16 S3 S5 S7	Explains how they select and engineer techniques to develop the machine learning solution and the processes used to identify variables and features that can impact stability of model performance during testing and when applying changes to existing models in the live environment. K8, S3 Explains feature engineering, selection and pre-processing and their importance in effective machine learning. K9 Describes deployment approaches for new data pipelines and automated processes and	Critically evaluates how their application of techniques ensures the validity and robustness of machine learning. K10, S7

	how they create and deploy models to produce machine learning solutions. K12, S5 Explains how programming languages, integrated development environments and modern machine learning libraries are used. K16 Explains the techniques applied for output model testing and tuning in order to access accuracy, fit, validity and robustness. K10, S7	
Model deployment K13 K31 S8 S10 S13 S14 S16 S20	Outlines the data and information security standard, ethical practices, policies and procedures relevant to data management. K13 Explains how they track, test and monitor continual learning models in to the live environment, ensuring that they remain fit for purpose and stable. S10, S13, S14 Explains how they transition prototypes into the live environment. S16 How they assess system vulnerabilities and mitigate the threats or risks to assets, data and cyber security. S8 Explains how they support the evaluation and validation of machine learning models and statistical evidence to minimise algorithmic bias being introduced using their knowledge of software development best practices. K31, S20	Justify how validation of machine learning models minimised bias and the impact this has on prototypes in the live environment. S16, S20

Collaborative working K22 K23 S28 S29	Explains how they create and use reports, presentations and other documentation on the model development to confirm stakeholder and end user engagement and approval for handover implementation. K22, S29 Explains how they produce and maintain technical documentation using their knowledge of machine learning and data science techniques to enhance the work of other members of the team and meet technical and non-technical user requirements. K23, S28	None
Sustainability K17 S15 S23 B2	Explains how they take responsibility for identifying, advocating and embedding ML solutions to deliver environmental and operational sustainable outcomes by considering principles that support organisational strategies and objectives for environmental sustainability. K17, S15, S23, B2	None

Professional discussion

Fail - does not meet pass criteria

THEME KSBS	PASS APPRENTICES MUST DEMONSTRATE ALL OF THE PASS DESCRIPTORS	DISTINCTION APPRENTICES MUST DEMONSTRATE ALL OF THE PASS DESCRIPTORS AND ALL OF THE DISTINCTION DESCRIPTORS
Engineering principles K1 K5 K18 K20 K30 S33 S34	Explains the purpose, methodologies and applications for machine learning and AI solutions. K1 Outlines the differences and applications of machine learning methods and models. K5 Describes the relationship between mathematical principles and core techniques in machine learning and data science within the organisational context. K18 Explains the sources of error and algorithmic bias and how they may be affected by choice of dataset and methodologies applied using practices. K20 Explains how they integrate AI based approaches into existing new processes and how they proactively identify the potential for automation. K30, S33, S34	None
Model testing and improvement K4 K14 K19 K25 S1 S9 S11 S12 B5	Articulates how they assess vulnerabilities to ensure security considerations throughout the development process and outlines the implications associated throughout at all stages of the machine learning lifecycle. K4, S1 Explains how they evaluate choices at each stage of the data process, taking the associated regulatory legal, ethical, governance and quality control issues into consideration. S11, K25 Summarises how they operate in settings of technical complexity	Critically evaluates why they refine the model to improve solution performance. S9, K14

	and uncertainty to evaluate software solutions and apply machine learning and data science techniques to solve complex business problems. K19, S12, B5 Outlines how they refine or re-engineer the model to improve solution performance whilst recording, logging and managing change using appropriate tools and documentation. K14, S9	
Model management K11 K15 S6 S17 S18 S19 S22 S24 S25	Explains how they identify and select platform architecture and hardware and apply machine learning methods to contribute to solving a computational problem and maximise the impact to the organisation. K11, S22 Outlines the implications of data types on security and scalability and the cost of local, remote or distributed solutions and how they consider the risks with using digital and physical supply chains. K15, S18, S19 Explains how they monitor model drift, data drift and performance metrics to ensure that systems are robust when moving outside of their domain of applicability. S24 Explains how they have developed a process to decommission assets and manage current and legacy models in line with policy and procedures. S25 Explains how they work in compliance with policies, governance, industry regulation and standards to complete audit activities whilst documenting the creation and operation and	Justifies how their consideration of risk and implications has impacted on the security and scalability of machine learning and AI infrastructure. K15 S19 Evaluates how systems are robust as a result of monitoring of data curation and controlling quality and performance of metrics. S22 S24

	management of assets during the model lifecycle. S6, S17	
Compliance and assurance K24 K26 K27 S32 B4	Explains the cyber security culture in an organisation and how it may contribute to security risk for Machine Learning solutions. K27 Explains how they apply machine learning principles, standards and assurance frameworks to ensure the safe interoperable use of data, machine learning and artificial intelligence. S32 Explains how they act with integrity in relation to the ethical aspects associated with the use of data and machine learning models, giving regard to legal, ethical and regulatory requirements. K26, B4, K24	None
Collaborative working K21 K29 S26 S27 S30 B3	Explains how they use strategies to engage with a diverse range of stakeholders, suppliers and multi-disciplinary teams to co-ordinate, negotiate and manage expectations and deal with conflicting priorities, interest and timescales. K21, S27 Describes how they act inclusively when collaborating with people from technical and non-technical backgrounds, contributing to knowledge sharing, management and empowerment across the broader team, whilst complying with equality, diversity and inclusion policies and procedures. S30, B3 Outlines how their own role supports the organisations strategy and objectives and how they work in unpredictable and	Justifies their strategies when working with stakeholders and how this positively impacts on the organisation. S27

	changing circumstances to make impartial decisions whilst respecting the opinions and views of others. K29, S26	
Continuous professional development K28 S31 B1	Explains how they use their initiative to identify new emerging technological developments and trends to ensure knowledge of machine learning and AI is up to date. K28, S31, B1	None

Overall EPA grading

Performance in the EPA determines the overall grade of:

- fail
- pass
- merit
- distinction

An independent assessor must individually grade the project evaluation report, presentation and questioning and professional discussion in line with this EPA plan.

The EPAO must combine the individual assessment method grades to determine the overall EPA grade.

If the apprentice fails one assessment method or more, they will be awarded an overall fail.

To achieve an overall pass, the apprentice must achieve at least a pass in all the assessment methods. Assessment grades follow the table included.

Grades from individual assessment methods must be combined in the following way to determine the grade of the EPA overall.

PROJECT EVALUATION REPORT, PRESENTATION AND QUESTIONING	PROFESSIONAL DISCUSSION	OVERALL GRADING
Fail	Fail	Fail
Any grade	Fail	Fail
Fail	Any grade	Fail
Pass	Pass	Pass
Distinction	Pass	Merit
Distinction	Distinction	Distinction
Pass	Distinction	Merit

Re-sits and re-takes

If the apprentice fails one assessment method or more, they can take a re-sit or a re-take at their employer's discretion. The apprentice's employer needs to agree that a re-sit or re-take is appropriate. A re-sit does not need further learning, whereas a re-take does. The apprentice should have a supportive action plan to prepare for a re-sit or a re-take.

The employer and the EPAO should agree the timescale for a re-sit or re-take. A re-sit is typically taken within 3 months of the EPA outcome notification. The timescale for a re-take is dependent on how much re-training is required and is typically taken within 6 months of the EPA outcome notification.

If the apprentice fails the project assessment method, they must amend the project output in line with the independent assessor's feedback. The apprentice will be given 2 weeks to rework and submit the amended report.

Failed assessment methods must be re-sat or re-taken within a 6-month period from the EPA outcome notification, otherwise the entire EPA will need to be re-sat or re-taken in full.

Re-sits and re-takes are not offered to an apprentice wishing to move from pass to a higher grade.

The apprentice will get a maximum EPA grade of if pass they need to re-sit or re-take one or more assessment methods, unless the EPAO determines there are exceptional circumstances.

Roles and responsibilities

ROLES	RESPONSIBILITIES
Apprentice	As a minimum, the apprentice should: • complete on-programme training to meet the KSBs as outlined in the apprenticeship standard for a minimum of 12 months • complete the required amount of off-the-job training specified by the apprenticeship funding rules and as arranged by the employer and training provider • understand the purpose and importance of EPA • prepare for and undertake the EPA including meeting all gateway requirements
Employer	As a minimum, the apprentice's employer must: • select the training provider • work with the training provider to select the EPAO • work with the training provider, where applicable, to support the apprentice in the workplace and to provide the opportunities for the apprentice to develop the KSBs • arrange and support off-the-job training to be undertaken by the apprentice • decide when the apprentice is working at or above the apprenticeship standard and is ready for EPA • ensure the apprentice is prepared for the EPA • ensure that all supporting evidence required at the gateway is submitted in line with this EPA plan • confirm arrangements with the EPAO for the EPA in a timely manner, including who, when, where • provide the EPAO with access to any employer-specific documentation as required for example, company policies • ensure that the EPA is scheduled with the EPAO for a date and time which allows appropriate opportunity for the apprentice to meet the KSBs • ensure the apprentice is given sufficient time away from regular duties to prepare for, and complete the EPA • ensure that any required supervision during the EPA period, as stated within this EPA plan, is in place • ensure the apprentice has access to the resources used to fulfil their role and carry out the EPA for workplace based assessments remain independent from the delivery of the EPA

	• pass the certificate to the apprentice upon receipt

EPAO	As a minimum, the EPAO must:
	
	• conform to the requirements of this EPA plan and deliver its requirements in a timely manner
	• conform to the requirements of the apprenticeship provider and assessment register
	• conform to the requirements of the external quality assurance provider (EQAP)
	• understand the apprenticeship including the occupational standard and EPA plan
	• make all necessary contractual arrangements including agreeing the price of the EPA
	• develop and produce assessment materials including specifications and marking materials, for example mark schemes, practice materials, training material
	• maintain and apply a policy for the declaration and management of conflict of interests and independence. This must ensure, as a minimum, there is no personal benefit or detriment for those delivering the EPA or from the result of an assessment. It must cover:
	  ◦ apprentices
	  ◦ employers
	  ◦ independent assessors
	  ◦ any other roles involved in delivery or grading of the EPA
	• have quality assurance systems and procedures that ensure fair, reliable and consistent assessment and maintain records of internal quality assurance (IQA) activity for external quality assurance (EQA) purposes
	• appoint independent, competent, and suitably qualified assessors in line with the requirements of this EPA plan
	• appoint administrators, invigilators and any other roles where required to facilitate the EPA
	• deliver induction, initial and on-going training for all their independent assessors and any other roles involved in the delivery or grading of the EPA as specified within this EPA plan. This should include how to record the rationale and evidence for grading decisions where required
	• conduct standardisation with all their independent assessors before allowing them to deliver an EPA, when the EPA is

	updated, and at least once a year • conduct moderation across all of their independent assessors' decisions once EPAs have started according to a sampling plan, with associated risk rating of independent assessors • monitor the performance of all their independent assessors and provide additional training where necessary • develop and provide assessment recording documentation to ensure a clear and auditable process is in place for providing assessment decisions and feedback to all relevant stakeholders • use language in the development and delivery of the EPA that is appropriate to the level of the apprenticeship • arrange for the EPA to take place in a timely manner, in consultation with the employer • provide information, advice, and guidance documentation to enable apprentices, employers and training providers to prepare for the EPA • confirm the gateway requirements have been met before they start the EPA for an apprentice • arrange a suitable venue for the EPA • maintain the security of the EPA including, but not limited to, verifying the identity of the apprentice, invigilation and security of materials • where the EPA plan permits assessment away from the workplace, ensure that the apprentice has access to the required resources and liaise with the employer to agree this if necessary • confirm the overall grade awarded • maintain and apply a policy for conducting appeals
Independent assessor	As a minimum, an independent assessor must: • be independent, with no conflict of interest with the apprentice, their employer or training provider, specifically, they must not receive a personal benefit or detriment from the result of the assessment • have, maintain and be able to evidence up-to-date knowledge and expertise of the occupation • have the competence to assess the EPA and meet the requirements of the IQA section of this EPA plan • understand the apprenticeship's occupational standard and EPA plan

	• attend induction and standardisation events before they conduct an EPA for the first time, when the EPA is updated, and at least once a year • use language in the delivery of the EPA that is appropriate to the level of the apprenticeship • work with other personnel, where used, in the preparation and delivery of assessment methods • conduct the EPA to assess the apprentice against the KSBs and in line with the EPA plan • make final grading decisions in line with this EPA plan • record and report assessment outcome decisions • comply with the IQA requirements of the EPAO • comply with external quality assurance (EQA) requirements
Training provider	As a minimum, the training provider must: • conform to the requirements of the apprenticeship provider and assessment register • ensure procedures are in place to mitigate against any conflict of interest • work with the employer and support the apprentice during the off-the-job training to provide the opportunities to develop the KSBs as outlined in the occupational standard • deliver training to the apprentice as outlined in their apprenticeship agreement • monitor the apprentice's progress during any training provider led on-programme learning • ensure the apprentice is prepared for the EPA • work with the employer to select the EPAO • advise the employer, upon request, on the apprentice's readiness for EPA • ensure that all supporting evidence required at the gateway is submitted in line with this EPA plan • remain independent from the delivery of the EPA

Reasonable adjustments

The EPAO must have reasonable adjustments arrangements for the EPA.

This should include:

- how an apprentice qualifies for a reasonable adjustment
- what reasonable adjustments may be made

Adjustments must maintain the validity, reliability and integrity of the EPA as outlined in this EPA plan.

Special considerations

The EPAO must have special consideration arrangements for the EPA.

This should include:

- how an apprentice qualifies for a special consideration
- what special considerations will be given

Special considerations must maintain the validity, reliability and integrity of the EPA as outlined in this EPA plan.

Internal quality assurance

Internal quality assurance refers to the strategies, policies and procedures that an EPAO must have in place to ensure valid, consistent and reliable EPA decisions.

EPAOs for this EPA must adhere to the requirements within the roles and responsibilities table.

They must also appoint independent assessors who:

- have recent relevant experience of the occupation or sector to at least occupational level 7 gained in the last 5 years or significant experience of the occupation or sector

Value for money

Affordability of the EPA will be aided by using at least some of the following:

- completing applicable assessment methods online, for example computer-based assessment
- utilising digital remote platforms to conduct applicable assessment methods
- using the employer's premises
- conducting assessment methods on the same day

Professional recognition

This apprenticeship aligns with:

- BCS, The Chartered Institute for IT for Advanced RITTech

KSB mapping table

KNOWLEDGE	ASSESSMENT METHODS
K1 The purpose, methodologies and applications for ML AI solutions such as Machine Learning, Computer (Machine) Vision, batched learning systems, Robotics, Generative Transformer Models and Natural & Large Language Processing (NLP and LLMs) Models.	Professional discussion
K2 The stages of the machine learning lifecycle. Including establishing the model objectives, data preparation, building and training the model, ML problem framing, testing and evaluating the model using the preferred framework, deploying the modelling and monitoring, maintaining and updating the model using process frameworks such as Quality Assurance and either online, continuous (CLS) or batched learning systems.	Project evaluation report, presentation and questioning
K3 Vulnerabilities related to confidentiality, authentication, non-repudiation, service integrity, network security, planned or unplanned adversarial danger, threat or attack, host OS security, physical security and the implications and preventative mitigations for these at all stages of the machine learning lifecycle.	Project evaluation report, presentation and questioning
K4 Project Management methodologies and techniques for machine learning activities such as CRISP-ML Cross Industry Standard Process.	Professional discussion
K5 Differences and applications of machine learning methods, and models such as: supervised learning; semi supervised learning; unsupervised learning; natural language processing ; reinforcement learning; ensemble learning; predictive using tools for experiment tracking, orchestration, versioning, deployment and monitoring.	Professional discussion
K6 The risks that might occur for example bias, security, quality or over fitting in the product lifecycle during building, testing and through to deployment of ML models in the live environment.	Project evaluation report, presentation and questioning

K7 How to identify and select the performance metrics of the proposed model in the context of the business need.	Project evaluation report, presentation and questioning
K8 The processes used to identify variables and features that can impact stability of model performance during testing and when applying changes to existing models in the live environment.	Project evaluation report, presentation and questioning
K9 The importance of feature engineering, selection and pre-processing in effective machine learning.	Project evaluation report, presentation and questioning
K10 Machine learning implementation principles for data engineering solutions including quality, security, efficiency, validity, training, testing and tuning.	Project evaluation report, presentation and questioning
K11 How machine learning methods are applied to maximise the impact to the organisation.	Professional discussion
K12 Deployment approaches for new data pipelines and automated processes.	Project evaluation report, presentation and questioning
K13 Data and information security standards, ethical practices, policies and procedures relevant to data management activities such as data lineage, data retention and metadata management.	Project evaluation report, presentation and questioning
K14 Change management processes for ML solutions; recording and logging change using appropriate tools and documentation.	Professional discussion
K15 The implications of data types (for example variety, quality, formats) on security, scalability, governance for ML and or AI infrastructure, and cost of local, remote or distributed solutions such as cloud and other SaaS and PasS ML/AI providers.	Professional discussion

K16 How to use programming languages, integrated development environments and modern machine learning libraries.	Project evaluation report, presentation and questioning
K17 Principles for engineering environmentally sustainable ML solutions, that support organisational strategies and objectives for environmental sustainability.	Project evaluation report, presentation and questioning
K18 The relationship between mathematical principles and core techniques in machine learning and data science within the organisational context.	Professional discussion
K19 How to solve problems and evaluate software solutions via analysis of test data including synthetic data and results from research, feasibility, acceptance and usability testing.	Professional discussion
K20 Sources of error and algorithmic bias, including how they may be affected by choice of dataset and methodologies applied using practices such as Explicability and Explainable AI (XAI).	Professional discussion
K21 The methods and techniques used to communicate concepts and messages to meet the needs of the audience, adapting communication techniques accordingly.	Professional discussion
K22 Approaches and strategies to stakeholder engagement including engagement with the end user	Project evaluation report, presentation and questioning
K23 How machine learning and data science techniques support and enhance the work of other members of the team.	Project evaluation report, presentation and questioning
K24 Concepts of data governance, including regulatory requirements, data privacy, security, trustworthiness and quality control.	Professional discussion
K25 Legislation, regulation, governance and guidance assurance frameworks for example AREA or SAFE D and their application to	Professional discussion

the safe interoperable use of data, machine learning and artificial intelligence.	
K26 The ethical aspects associated with the use and collation of data and machine learning models.	Professional discussion
K27 What the cyber security culture in an organisation is, and how it may contribute to security risk.	Professional discussion
K28 How to identify trends and emerging technologies to ensure knowledge is up to date with new developments in machine learning and AI such as AI embedded within tooling.	Professional discussion
K29 How own role supports ML solutions in accordance with organisational strategies, business requirements, Corporate Governance Principles, Social Corporate Responsibilities, legal regulations and Ethical Practices.	Professional discussion
K30 AI based approaches, including those provided by third-party vendors' (Application Programming Interfaces), into existing and new processes.	Professional discussion
K31 Software development best practices; for example, software testing, version control, continuous integration and continuous delivery.	Project evaluation report, presentation and questioning

SKILL	ASSESSMENT METHODS
S1 Assess vulnerabilities of the proposed design, to ensure that security considerations are built in from inception and throughout the development process.	Professional discussion
S2 Translate business needs and technical problems to scope machine learning engineering solutions.	Project evaluation report, presentation and questioning
S3 Select and engineer data sets, algorithms and modelling techniques required to develop the machine learning solution.	Project evaluation report, presentation and questioning
S4 Apply methodologies and project management techniques for the machine learning activities.	Project evaluation report, presentation and questioning
S5 Create and deploy models to produce machine learning solutions.	Project evaluation report, presentation and questioning
S6 Document the creation, operation and lifecycle management of assets during the model lifecycle.	Professional discussion
S7 Apply techniques for output model testing and tuning to assess accuracy, fit, validity and robustness.	Project evaluation report, presentation and questioning
S8 Assess system vulnerabilities and mitigate the threats or risks to assets, data and cyber security.	Project evaluation report, presentation and questioning
S9 Refine or re-engineer the model to improve solution performance.	Professional discussion
S10 Apply techniques for monitoring models in the live environment to check they remain fit for purpose and stable.	Project evaluation report, presentation and questioning
S11	Professional discussion

Consider the associated regulatory, legal, ethical and governance issues when evaluating choices at each stage of the data process.	
S12 Apply machine learning and data science techniques to solve complex business problems.	Professional discussion
S13 Track and test continual learning models.	Project evaluation report, presentation and questioning
S14 Analyse test data, interpret results and evaluate the suitability of proposed solutions both new and inherited models, considering current and future business requirements.	Project evaluation report, presentation and questioning
S15 Identify, consider and advocate for ML solutions to deliver an environmental and operational sustainable outcome.	Project evaluation report, presentation and questioning
S16 Transition prototypes into the live environment.	Project evaluation report, presentation and questioning
S17 Complete audit activities in compliance with policies, governance, industry regulation and standards.	Professional discussion
S18 Consider the risks with using digital and physical supply chains.	Professional discussion
S19 Ensure the model capacity is scaled in proportion to the operating requirements.	Professional discussion
S20 Support the evaluation and validation of machine learning models and statistical evidence to minimise algorithmic bias being introduced.	Project evaluation report, presentation and questioning
S21 Monitor data curation and data quality controls including for synthetic data.	Project evaluation report, presentation and questioning

S22 Identify and select the machine learning or artificial intelligence platform architecture and specific hardware, to contribute to solving a computational problem using allocated resources.	Professional discussion
S23 Identify and embed changes in work to deliver sustainable outcomes.	Project evaluation report, presentation and questioning
S24 Monitor model data drift, using performance metrics to ensure systems are robust when moving outside of their domain of applicability.	Professional discussion
S25 Develop a process to decommission assets in line with policy and procedures. Manage current and legacy models in line with industry approaches.	Professional discussion
S26 Undertake independent, impartial decision-making respecting the opinions and views of others in complex, unpredictable and changing circumstances.	Professional discussion
S27 Coordinate, negotiate with and manage expectations of diverse stakeholders suppliers and multi-disciplinary teams with conflicting priorities, interests and timescales.	Professional discussion
S28 Produce and maintain technical documentation explaining the data product, that meets organisational, technical and non-technical user requirements, retaining critical information.	Project evaluation report, presentation and questioning
S29 Create and disseminate reports, presentations and other documentation that details the model development to confirm stakeholder approval for handover to implementation.	Project evaluation report, presentation and questioning
S30 Comply with equality, diversity, and inclusion policies and procedures in the workplace.	Professional discussion
S31	Professional discussion

Horizon scan to identify new technological developments that offer increased performance of data products.	
S32 Apply Machine Learning principles and standards such as, organisational policies, procedures or professional body requirements.	Professional discussion
S33 Integrate AI-based approaches, including those provided by third-party vendors' Application Programming Interfaces, into existing and new processes.	Professional discussion
S34 Proactive identification of the potential for automation for example through AI solutions embedded within tooling.	Professional discussion

BEHAVIOUR	ASSESSMENT METHODS
B1 Uses initiative and innovation concerning new and emerging technologies through self directed learning and horizon scanning.	Professional discussion
B2 Takes personal responsibility and prioritises sustainable outcomes in how they carry out the duties of their role.	Project evaluation report, presentation and questioning
B3 Acts inclusively when collaborating with people from technical and non-technical backgrounds. Contributing to knowledge sharing, management and empowerment across the broader team.	Professional discussion
B4 Acts with integrity, giving due regard to legal, ethical and regulatory requirements.	Professional discussion
B5 Operates in settings of technical complexity and uncertainty.	Professional discussion

Mapping of KSBs to grade themes

Project evaluation report, presentation and questioning

KSBS GROUPED BY THEME	KNOWLEDGE	SKILLS	BEHAVIOUR
Model scoping K2 K3 K6 K7 S2 S4 S21	The stages of the machine learning lifecycle. Including establishing the model objectives, data preparation, building and training the model, ML problem framing, testing and evaluating the model using the preferred framework, deploying the modelling and monitoring, maintaining and updating the model using process frameworks such as Quality Assurance and either online, continuous (CLS) or batched learning systems. (K2) Vulnerabilities related to confidentiality, authentication, non-repudiation, service integrity, network security, planned or unplanned adversarial danger, threat or attack, host OS security, physical security and the implications and preventative mitigations for these at all stages of the machine learning lifecycle. (K3) The risks that might occur for example bias, security, quality or over fitting in the	Translate business needs and technical problems to scope machine learning engineering solutions. (S2) Apply methodologies and project management techniques for the machine learning activities. (S4) Monitor data curation and data quality controls including for synthetic data. (S21)	None

	product lifecycle during building, testing and through to deployment of ML models in the live environment. (K6) How to identify and select the performance metrics of the proposed model in the context of the business need. (K7)		
Experiment and tracking K8 K9 K10 K12 K16 S3 S5 S7	The processes used to identify variables and features that can impact stability of model performance during testing and when applying changes to existing models in the live environment. (K8) The importance of feature engineering, selection and pre-processing in effective machine learning. (K9) Machine learning implementation principles for data engineering solutions including quality, security, efficiency, validity, training, testing and tuning. (K10) Deployment approaches for new data pipelines and automated processes. (K12)	Select and engineer data sets, algorithms and modelling techniques required to develop the machine learning solution. (S3) Create and deploy models to produce machine learning solutions. (S5) Apply techniques for output model testing and tuning to assess accuracy, fit, validity and robustness. (S7)	None

	How to use programming languages, integrated development environments and modern machine learning libraries. (K16)		

Model deployment K13 K31 S8 S10 S13 S14 S16 S20	Data and information security standards, ethical practices, policies and procedures relevant to data management activities such as data lineage, data retention and metadata management. (K13) Software development best practices; for example, software testing, version control, continuous integration and continuous delivery. (K31)	Assess system vulnerabilities and mitigate the threats or risks to assets, data and cyber security. (S8) Apply techniques for monitoring models in the live environment to check they remain fit for purpose and stable. (S10) Track and test continual learning models. (S13) Analyse test data, interpret results and evaluate the suitability of proposed solutions both new and inherited models, considering current and future business requirements. (S14) Transition prototypes into the live environment. (S16) Support the evaluation and validation of machine learning models and statistical evidence to minimise algorithmic bias being introduced. (S20)	None
Collaborative working K22 K23 S28 S29	Approaches and strategies to stakeholder engagement including engagement with the end user (K22)	Produce and maintain technical documentation explaining the data product, that meets organisational, technical and non-	None

	How machine learning and data science techniques support and enhance the work of other members of the team. (K23)	technical user requirements, retaining critical information. (S28) Create and disseminate reports, presentations and other documentation that details the model development to confirm stakeholder approval for handover to implementation. (S29)	
Sustainability K17 S15 S23 B2	Principles for engineering environmentally sustainable ML solutions, that support organisational strategies and objectives for environmental sustainability. (K17)	Identify, consider and advocate for ML solutions to deliver an environmental and operational sustainable outcome. (S15) Identify and embed changes in work to deliver sustainable outcomes. (S23)	Takes personal responsibility and prioritises sustainable outcomes in how they carry out the duties of their role. (B2)

Professional discussion

KSBS GROUPED BY THEME	KNOWLEDGE	SKILLS	BEHAVIOUR
Engineering principles K1 K5 K18 K20 K30 S33 S34	The purpose, methodologies and applications for ML AI solutions such as Machine Learning, Computer (Machine) Vision, batched learning systems, Robotics, Generative Transformer Models and Natural & Large Language Processing (NLP and LLMs) Models. (K1) Differences and applications of machine learning methods, and models such as: supervised learning; semi supervised learning; unsupervised learning; natural language processing ; reinforcement learning; ensemble learning; predictive using tools for experiment tracking, orchestration, versioning, deployment and monitoring. (K5) The relationship between mathematical principles and core techniques in machine learning and data science within the organisational context. (K18) Sources of error and algorithmic bias,	Integrate AI-based approaches, including those provided by third-party vendors' Application Programming Interfaces, into existing and new processes. (S33) Proactive identification of the potential for automation for example through AI solutions embedded within tooling. (S34)	None

	including how they may be affected by choice of dataset and methodologies applied using practices such as Explicability and Explainable AI (XAI). (K20) AI based approaches, including those provided by third-party vendors' (Application Programming Interfaces), into existing and new processes. (K30)		
Model testing and improvement K4 K14 K19 K25 S1 S9 S11 S12 B5	Project Management methodologies and techniques for machine learning activities such as CRISP-ML Cross Industry Standard Process. (K4) Change management processes for ML solutions; recording and logging change using appropriate tools and documentation. (K14) How to solve problems and evaluate software solutions via analysis of test data including synthetic data and results from research, feasibility, acceptance and usability testing. (K19)	Assess vulnerabilities of the proposed design, to ensure that security considerations are built in from inception and throughout the development process. (S1) Refine or re-engineer the model to improve solution performance. (S9) Consider the associated regulatory, legal, ethical and governance issues when evaluating choices at each stage of the data process. (S11) Apply machine learning and data science techniques to solve complex	Operates in settings of technical complexity and uncertainty. (B5)

	Legislation, regulation, governance and guidance assurance frameworks for example AREA or SAFE D and their application to the safe interoperable use of data, machine learning and artificial intelligence. (K25)	business problems. (S12)	
Model management K11 K15 S6 S17 S18 S19 S22 S24 S25	How machine learning methods are applied to maximise the impact to the organisation. (K11) The implications of data types (for example variety, quality, formats) on security, scalability, governance for ML and or AI infrastructure, and cost of local, remote or distributed solutions such as cloud and other SaaS and PasS ML/AI providers. (K15)	Document the creation, operation and lifecycle management of assets during the model lifecycle. (S6) Complete audit activities in compliance with policies, governance, industry regulation and standards. (S17) Consider the risks with using digital and physical supply chains. (S18) Ensure the model capacity is scaled in proportion to the operating requirements. (S19) Identify and select the machine learning or artificial intelligence platform architecture and specific hardware, to contribute to solving a computational problem using allocated resources. (S22)	None

		Monitor model data drift, using performance metrics to ensure systems are robust when moving outside of their domain of applicability. (S24) Develop a process to decommission assets in line with policy and procedures. Manage current and legacy models in line with industry approaches. (S25)	
Compliance and assurance K24 K26 K27 S32 B4	Concepts of data governance, including regulatory requirements, data privacy, security, trustworthiness and quality control. (K24) The ethical aspects associated with the use and collation of data and machine learning models. (K26) What the cyber security culture in an organisation is, and how it may contribute to security risk. (K27)	Apply Machine Learning principles and standards such as, organisational policies, procedures or professional body requirements. (S32)	Acts with integrity, giving due regard to legal, ethical and regulatory requirements. (B4)
Collaborative working K21 K29 S26 S27 S30 B3	The methods and techniques used to communicate concepts and messages to meet the needs of the audience, adapting communication techniques accordingly. (K21)	Undertake independent, impartial decision-making respecting the opinions and views of others in complex, unpredictable and changing circumstances. (S26)	Acts inclusively when collaborating with people from technical and non-technical backgrounds. Contributing to knowledge sharing, management and empowerment across

	How own role supports ML solutions in accordance with organisational strategies, business requirements, Corporate Governance Principles, Social Corporate Responsibilities, legal regulations and Ethical Practices. (K29)	Coordinate, negotiate with and manage expectations of diverse stakeholders suppliers and multi-disciplinary teams with conflicting priorities, interests and timescales. (S27) Comply with equality, diversity, and inclusion policies and procedures in the workplace. (S30)	the broader team. (B3)
Continuous professional development K28 S31 B1	How to identify trends and emerging technologies to ensure knowledge is up to date with new developments in machine learning and AI such as AI embedded within tooling. (K28)	Horizon scan to identify new technological developments that offer increased performance of data products. (S31)	Uses initiative and innovation concerning new and emerging technologies through self directed learning and horizon scanning. (B1)

Version log

Version	Change detail	Earliest start date	Latest start date
Revised version awaiting implementation	Assessment plan revised	11/12/2026	Not set
1.0	Approved for delivery	18/12/2024	10/12/2026